

Guest Lectures

1. "Recombinant-DNA technology" Guest lecture in BIOL:3774 (Molecular Biology, 3 credits). 2/20/98. Virginia Tech, Blacksburg VA.
2. "Molecular mechanisms of neurons and neural networks". Guest lecture in BIOL:3124 (Cell Physiology, 3 credits). 4/28/97. Virginia Tech, Blacksburg, VA.
3. "Molecular tools used to dissect cellular mechanisms". Guest lecture in BIOL:3774 (Molecular Biology, 3 credits). 3/26/97. Virginia Tech, Blacksburg VA.
4. "A historical and contemporary perspective of bacteriophage genetics" Guest lecture in BIOL:4634 (Microbial Genetics, 3 credits). 4/3/96. Virginia Tech, Blacksburg VA.

Oral Presentations

1. "Analysis of the Arabidopsis thaliana 2YY6 chalcone synthase allele". Plant Molecular Biology Group. 12/8/94. Virginia Tech, Blacksburg, VA.
2. "Flavonoid-independent fertility in Arabidopsis". Botany seminar series. 9/8/95. Virginia Tech, Blacksburg, VA.
3. "Analysis of an Arabidopsis chalcone synthase mutation: evidence for flavonoid-independent fertility". 6th International conference on Arabidopsis research. 6/7-11/95. Madison, Wisconsin.
4. "Assembly of malate dehydrogenase and citrate synthase into a model enzyme complex". Plant Molecular Biology Group. 3/3/96. Virginia Tech, Blacksburg VA.
5. "Analysis of flavonoid metabolism in Arabidopsis". 75th Annual meeting of the Virginia Academy of Sciences. 5/23/97. Virginia Tech, Blacksburg, VA.
6. "Quantitation of protein and small molecule analytes at extremely sensitive levels". Banbury workshop on designer molecules for biosensor applications. August 12-14, 2002. Cold Spring Harbor Laboratory, NY.
7. "Novel proteomics technology to monitor protein identity and abundance in complex mixtures". First annual Alpha project research symposium. December 13, 2002, Berkeley, CA.
8. "Engineering and use of protein-DNA fusions". Second annual Alpha project research symposium. June 20, 2004, Berkeley, CA.
9. "Engineering and use of protein-DNA fusions to detect and count small numbers of molecules". Bay Area Bioinformatics Group. July 22, 2004. Stanford University, Palo Alto, CA.
10. "Use of protein-DNA chimera to quantify biological molecules". Second Annual Alpha Factor retreat. Costanoa Resort, Pescadero, CA.
11. "Creation and use of highly sensitive measurement technologies". United States Department of Agriculture Seminal Series. February 11, 2005. Western Regional Research Center. Albany, CA.
12. "Using protein-DNA chimeras to detect and count small numbers of biological molecules". Houghton College Interdisciplinary Seminar. April 8, 2005. Houghton College, Houghton, NY.
13. "Construction and use of protein-DNA fusions to quantify biological molecules". October 25, 2005. Genospectra, Inc. Fremont, CA.
14. "Construction and use of protein-DNA fusions to quantify biological molecules". November 14, 2005. Lawrence Livermore National Lab. Livermore, CA.
15. "Disassembling and counting the parts of yeast". January 28, 2006. CEGS Alpha project retreat, Santa Cruz, CA.

16. "Using Protein-DNA Chimeras to Detect and Count Small Numbers of Molecules". February 7, 2006. Canary Fund Technical Advisory Board Meeting, Santa Clara, CA.

B. Selected peer-reviewed publications (in chronological order). Do not include publications submitted or in preparation.

1. Burbulis, I.E., Iacobucci, M., Shirley, B.W. "A null mutation in the first enzyme of flavonoid biosynthesis does not affect male fertility in Arabidopsis". The Plant Cell. Vol.8, 1013-1025, June 1996.
2. Burbulis, I.E., Pelletier, M.K., Cain, C.C., and Shirley, B.W. "Are flavonoids synthesized by a multi-enzyme complex?". Bulletin of the Southern Association of Agricultural Scientists in Biochemistry and Biotechnology. Vol.9, 29-36, 1996.
3. Burbulis, I.E. and Shirley, B.W. "Interactions among enzymes of the Arabidopsis flavonoid biosynthetic pathway". Proc. Natl. Acad. Sci. Vol. 96, 12929-12934, 1999.
4. Pelletier, M.K., Burbulis, I.E., Winkel-Shirley, B. "Disruption of specific flavonoid genes enhances the accumulation of flavonoid enzymes and end-products in Arabidopsis seedlings". Plant molecular Biology Vol. 40, 45-54, 1999.
5. Burbulis, I., Yamaguchi, K., Carlson, R., and Brent, R. "Using protein-DNA chimeras to detect and count small numbers of molecules". Nature Methods 2005, 2:31-37.
6. Brent, R. and Burbulis, I. (2005) Reply to "Lamp for tadpoles". Nature Methods 2005, 2:635-6.

C. Research Support. List selected ongoing or completed (during the last three years) research projects (federal and non-federal support). Begin with the projects that are most relevant to the research proposed in this application. Briefly indicate the overall goals of the projects and your role (e.g. PI, Co-Investigator, Consultant) in the research project. Do not list award amounts or percent effort in projects.

Ongoing Research Support

Canary Foundation

Title: "Tadpole assays for proteomic detection of early cancer"

The goal of this work is to develop tadpole technology so that it can be used to quantify ovarian cancer marker proteins that may be present in small amounts in serum. We believe that if tadpole assays are developed to quantify molecules from clinical samples, their relative simplicity, low cost, great sensitivity, accuracy, and large dynamic range will make them invaluable in workhorse detection of protein marker and/or proteomic molecular signatures of cancer.

Role: Co-Investigator

1 R33 CA114306-0A1 Brent (PI)

02/23/06-11/30/08

Title: "Tadpole Assays for the Molecular Assessment of Cancer"

The goal of this work is to use tadpole technology to detect and count small numbers of cancer marker molecules. This funding will be used to develop and "harden" these assays to the point that they can be tested in clinical cancer diagnosis.

Role: Co-Principal Investigator

Completed Research Support

Grant # 0301027. National Science Foundation. 10/01/02 - 4/30/03

Program in EIA-Revolutionary Computing.

Title: "Developing high sensitivity epitope detection reagents for field effect devices".

Goals: Engineering affinity proteins that bind Bacillus anthracis spores and toxin in vitro for creating field effects device sensors.

Role: Co-Principal Investigator